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ORIGINAL PAPER

Genome-wide Classification, Identification and Expression Profile of the C3HC4-type RING Finger Gene Family in Poplar (*Populus trichocarpa*)

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Abstract C3HC4-type RING finger genes belong to a large gene family in plants and play crucial roles in diverse stress responses. In this study, a comprehensive analysis of the poplar (*Populus trichocarpa*) C3HC4-type RING finger gene family was performed, including analysis of phylogenetic relationships, chromosomal locations, gene structures, motif compositions, possible cis-elements, and expression profiles. Ninety-one poplar C3HC4-type RING finger genes were identified, which were clustered phylogenetically into seven distinct subfamilies and distributed predominantly across 17 linkage groups. Motif analysis revealed the presence of 15 conserved motifs with unknown functions within the predicted proteins. Promoter cis-element analysis indicated that most of the C3HC4-type RING finger genes contained abiotic stress-related cis-elements. Further, expression profiling showed that a number of genes were expressed differentially across disparate tissues and under various stresses, such as wood formation and drought. Additionally, quantitative real-time RT-PCR was performed on 24 selected C3HC4-type RING finger genes to confirm their responses to drought stress treatment. We found that *PtrRHC7*, *20* and *70* were not only abundantly transcribed in leaves, but also were upregulated in roots under drought stress. These results represent the foundation for further investigation of the roles of these candidate genes and for future genetic engineering and gene functional studies in *Populus*.

Keywords *Populus trichocarpa* · C3HC4-type RING finger · Phylogenetics · Expression profile · Drought

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Introduction

Zinc-finger transcription factors (TF), one of the largest TF families in plants, play important roles in various cellular functions and multiple biological processes, such as morphogenesis, transcriptional activation, and environmental stress responses (Laity et al. 2001). The really interesting new gene (RING) finger belongs to the family of zinc finger domains first identified as a DNA-binding motif in the transcription factor TFIIIA from *Xenopus laevis* and comprises 40–60 amino acids that bind two atoms of zinc (Lovering et al. 1993). In addition to DNA binding, it also binds RNA, proteins, or lipid substrates (Berg and Shi 1996). Many proteins that harbor RING domain(s) are crucial for plant development and cellular processes, including gametogenesis, transcription regulation and signal transduction. Functions attributed to the RING domain itself include protein–protein interactions and ubiquitination (Borden and Freemont 1996). The majority of the canonical and modified *Arabidopsis* RING domain-containing proteins show a capacity for polyubiquitination with an E2 ubiquitin-conjugating enzyme (Stone et al. 2005). Additionally, most RING finger proteins are E3 ubiquitin ligases that mediate the transfer of ubiquitin to target proteins and play important roles in diverse aspects of cellular regulations in plants (Callis and Vierstra 2000).

According to the type of cysteine (Cys) and histidine (His) residue combination, the RING finger domain can be classified as either a canonical or modified RING finger. The canonical RING-finger protein family can be subcategorized into two classes: RING-H2 (C3H2C3) and RING-HC (C3HC4) (Kosarev et al. 2002). The consensus sequence of the C3HC4-type RING finger is Cys-X2-Cys-X(9-39)-Cys-X(1-3)-His-X(2-3)-Cys-X2-Cys-X(4-48)-Cys-X2-Cys, where X can be any amino acid, where phenylalanine and proline residues are highly conserved but not invariable, and the loops vary in length. However, other characterized RING domains contain a histidine (His) at metal ligand position 4, the C3HC4 type differs by the presence of a Cys residue at metal ligand position 5 (Lovering et al. 1993). Since its initial identification, many C3HC4-type RING finger proteins have been isolated from a variety of organisms, including animals, plants and viruses, and have been shown to possess many biological functions (Saurin et al. 1996).

The C3HC4-type RING finger proteins have been studied on a genomic scale in *Arabidopsis* and rice. A recent report showed that, among the 1,400 predicted E3 ubiquitin ligases, 469 proteins contained 477 RING domains, making them the third-largest gene family in *Arabidopsis*. Of these 477 RING domains, 186 RING-HC type domains were identified based on the typical RING domain structure (Stone et al. 2005). Some of the biological functions of RING-HC proteins in *Arabidopsis* have been predicted or deduced, including chromatin remodeling in seed dormancy (Liu et al. 2007a), photomorphogenesis (Hardtke et al. 2002), secretory pathways (Matsuda et al. 2001), root development (Wang et al. 2006), and regulation of drought stress-responsive gene expression (Qin et al. 2008). In rice, 119 RING-HC type domains were detected in the genome (Lim et al. 2010). The biological functions of some C3HC4-type RING finger genes have been suggestive of increased tolerance to abiotic stresses (Liu et al. 2007b) and defense response against pathogen invasion (Cheung et al. 2007).

Poplar (*Populus trichocarpa*) is used worldwide to produce a large variety of paper-based and timber products, making it a valuable forest resource and an important ecological species (Taylor 2002). Studies of the genus *Populus* have been a focus of attention for several years because of their economic importance in pulp and biofuel production. Completion of the *P. trichocarpa* genome sequence in 2006 has made it a model tree for other tree species (Tuskan et al. 2006). Compared with the extensive studies of C3HC4-type RING finger genes in many other plant species, little research on this gene family has been conducted previously in *Populus*. In this study, we performed a genome-wide analysis of the C3HC4-type RING finger gene family in *P. trichocarpa*. The expression profiles of this gene family in different tissues and organs of *Populus*, and under biotic and abiotic stress conditions, were analyzed using data from microarray and quantitative real-time RT-PCR (RT-qPCR) analyses. Our preliminary results represent a foundation for further investigation of the roles of these candidate genes and for future genetic engineering and gene functional studies in *Populus*.

Materials and Methods

Identification and Characteristics of the C3HC4-type RING Finger Gene Family

The protein database, genomic library and cDNA library of *P. trichocarpa* were downloaded from the Phytozome v9.1 (http://www.phytozome.net/search.php) and NCBI (http://www.ncbi.nlm.nih.gov/) databases. The Hidden Markov Model (HMM) profile of the C3HC4-type RING finger gene family (PF00097) was downloaded from the Protein family (Pfam 27.0) database (http://pfam.sanger.ac.uk/) (Finn et al. 2010). All of the located sequences were analyzed further manually to confirm the presence of a C3HC4-type RING finger domain using the SMART (http://smart.embl-heidelberg.de/) database (Letunic et al. 2004). WoLF PSORT (http://wolfpsort.org/) (Horton et al. 2007) was used to predict the subcellular localization of C3HC4-type RING finger proteins. The ExPasy site (http://web.expasy.org/protparam/) (Gasteiger et al. 2005) was used to calculate molecular weight and isoelectric point (pI) of the deduced polypeptides.

Phylogenetic Analysis

Multiple sequence alignment of the full-length protein sequences was performed with the Clustal X (version 1.83) program (Thompson et al. 1997) and adjusted manually using BioEdit 7.1 software (Hall 1999). Phylogenetic analysis using the neighbor-joining method and a bootstrap test carried out with 1,000 iterations (Tamura et al. 2007) were performed to construct unrooted phylogenetic trees with MEGA 5.0.

Chromosomal Location

Genes were mapped on chromosomes by identifying their chromosomal locations provided in the PopGenIE v3 database (http://www.popgenie.org/). Genes separated by five or fewer gene loci within a genetic distance of 100 kb were considered to be tandem duplicates. A schematic view of the reorganization of homologous chromosome segments was based on the most recent account of whole-genome duplication in *P. trichocarpa* (Tuskan et al. 2006).

Exon/Intron Structure and Conserved Motif Distribution

The Gene Structure Display Server 1.0 (Guo et al. 2007) was used to generate the exon/intron organization online. The multiple expectation maximization for motif elucidation (MEME) system (version 4.9) was used to identify conserved motifs for each C3HC4-type RING finger gene (Bailey et al. 2006).

Promoter cis-Elements Analysis

Promoter sequences (2 kb upstream of the translation start site) of all C3HC4-type RING finger genes were obtained from the NCBI database and analyzed using the PlantCARE database (Lescot et al. 2002).

exNorthern, exPlot and HeatMap

The expression profile for each gene was obtained by evaluating its expressed sequence tag (EST) representation among 17 cDNA libraries derived from different tissues and/or developmental stages using the exNorthern tool of the PopGenIE v3 database (http://www.popgenie.org/) (Sjodin et al. 2009). The 17 libraries were derived from several taxa of *Populus* (Street et al. 2008). The exPlot tool at PopGenIE was used to obtain the expression profiles of *Populus* C3HC4-type RING finger genes under different biotic stresses. Detailed description of C3HC4-type RING finger gene expression in response to different biotic stresses was downloaded from Sample List (http://popgenie.org/content/experiment-search; Table S1). The HeatMap tool at PopGenIE was used to visualize the heatmap of C3HC4-type RING finger genes across various developmental stages and under different stress conditions.

Plant Material Collection

For expression pattern analysis of the C3HC4-type RING finger gene family under drought stress, clonally propagated 7-month-old *P. trichocarpa* plants were cultured in half-strength Murashige and Skoog medium supplemented with 150 mM mannitol for the times indicated. Three biological replicates were performed for each stress treatment. Young leaves and roots from three different plants were collected at different time points (0, 1, 3, 6, 12, 24, and 48 h) after treatment. Three biological replicates were performed for each stress treatment. Each experiment was repeated at least twice with independent sample preparation to obtain reproducible results. All samples were immediately frozen in liquid nitrogen and stored at −80 °C until analysis.

RNA Isolation and RT-qPCR Verification

The cetyltrimethylammonium bromide method with minor modifications (Chang et al. 1993) was used to extract total RNA from roots and leaves. First-strand cDNA synthesis was carried out with the ReverTra Ace qPCR RT Master Mix with a gDNA Remover Kit (TOYOBO, Osaka, Japan) in accordance with the manufacturer's instructions. Primer Premier 5 was used to design primers with melting temperatures of 55–60 °C, primer lengths of 17–22 bp, and amplicon lengths of 110–220 bp. Details of the primers are shown in Table S2. SYBR Premix Ex Taq II (TaKaRa, Dalian, China) was used to perform RT-qPCR in accordance with the manufacturer's instructions. Reactions were prepared in a total volume of 20 μL containing 10 μL 2× SYBR Premix, 2 μL cDNA template, 6 μL ddH_2O, and 1 μL each primer to make a final concentration of 10 μM. The *P. trichocarpa* actin gene (GenBank ID: XM_002298674) was used as a reference gene. The PCR conditions and relative gene expression calculations were as previously described (Wang et al. 2013).

Results and Discussion

Identification of C3HC4-type RING Finger Genes in Populus

The profile hidden Markov model (HMM) of the Pfam C3HC4-type RING finger domain (PF00097) was exploited as the query to identify the C3HC4-type RING finger genes in the *Populus trichocarpa* genome. A total of 118 candidate C3HC4-type RING finger genes was identified in the *Populus* genome. All C3HC4-type RING finger candidates were analyzed manually using the SMART database to verify the presence of the C3HC4-type RING finger domain. Finally, a total of 91 C3HC4-type RING finger genes were identified. This number was less than the number present in the *Arabidopsis* and rice genomes (186 and 119, respectively) (Stone et al. 2005; Lim et al. 2010). All 91 genes were designated systematically from *PtrRHC1* to *PtrRHC91* in this study.

The *P. trichocarpa* C3HC4-type RING finger genes identified had a molecular weight ranging from 14,885.9 to 123,363.0 Da. The encoded proteins varied from 133 to 1,109 amino acids (aa) in length, with an average of 370 aa. The pI of the predicted proteins varied; for example, PtrRHC86 had a pI of 9.79, whereas that of PtrRHC17 was 3.83. WoLF PSORT was used to predict the location of the predicted proteins in the plant cell; 39 C3HC4-type RING finger proteins were predicted to be nuclear proteins, whereas 14 proteins were predicted to be cytoplasmic, while 13 were predicted as plasma membrane proteins, 17 as chloroplast proteins, 6 as extracellular matrix proteins, and 2 as endoplasmic reticulum proteins. Detailed information for the *PtrRHC* genes in *Populus* is provided in Table 1. Based on the WoLF PSORT analyses, we were able to roughly determine the gene position but experimental verification is required for more accurate localization.

Phylogenetic Analysis of C3HC4-type RING Finger Genes in Populus

To examine the phylogenetic relationships among the C3HC4-type RING finger domain proteins in *Populus*,

Table 1 C3HC4-type RING finger gene family in *Populus trichocarpa*

Gene name	Accession number	NCBI locus ID	Length (aa)	MW (Da)	pI	Localization[a]
PtrRHC1	POPTR_0013s11590	XM_006376215.1	461	49,711.7	9.17	plas
PtrRHC2	POPTR_0001s36840	XM_002300299.1	1,029	114,793.0	8.04	nucl
PtrRHC3	POPTR_0014s16830	XM_006375522.1	153	17,162.6	4.82	chlo
PtrRHC4	POPTR_0005s02460	XM_006382410.1	431	46,342.6	6.38	nucl
PtrRHC5	POPTR_0003s16080	XM_002303710.2	705	78,894.4	8.57	nucl
PtrRHC6	POPTR_0008s18980	XM_002311726.2	139	15,602.6	5.68	extr
PtrRHC7	POPTR_0011s07110	XM_002317366.2	378	40,264.8	4.70	cyto
PtrRHC8	POPTR_0004s10560	XM_002305934.2	531	59,356.7	6.63	nucl
PtrRHC9	POPTR_0011s09950	XM_002317334.2	551	60,527.4	6.11	nucl
PtrRHC10	POPTR_0001s38280	XM_002298862.2	551	60,707.6	6.20	nucl
PtrRHC11	POPTR_0006s15970	XM_002309203.2	316	35,838.3	5.27	ER
PtrRHC12	POPTR_0012s05000	XM_002318499.2	389	43,336.0	8.71	nucl
PtrRHC13	POPTR_0018s12240	XM_002324536.2	198	22,564.6	9.44	nucl
PtrRHC14	POPTR_0002s17150	XM_002302591.2	373	41,097.2	6.44	plas
PtrRHC15	POPTR_0002s14230	XM_002302474.2	879	100,792.1	6.48	nucl
PtrRHC16	POPTR_0011s15220	XM_002317003.2	253	27,456.0	5.88	plas
PtrRHC17	POPTR_0007s07600	XM_002310033.2	322	34,814.4	3.83	nucl
PtrRHC18	POPTR_0009s03970	XM_002313385.2	343	38,250.3	5.61	nucl
PtrRHC19	POPTR_0010s10240	XM_002314683.2	422	45,956.0	5.45	chlo
PtrRHC20	POPTR_0002s01890	XM_002301915.2	165	17,862.7	5.06	chlo
PtrRHC21	POPTR_0006s20810	XM_002309262.2	448	49,747.1	5.75	nucl
PtrRHC22	POPTR_0018s10720	XM_002325055.2	170	19,419.7	9.63	cyto
PtrRHC23	POPTR_0009s10230	XM_002313092.2	1,109	123,363.0	5.53	plas
PtrRHC24	POPTR_0018s06490	XM_006371875.1	309	35,093.6	5.87	cyto
PtrRHC25	POPTR_0008s08690	XM_002311269.2	225	25,241.9	4.77	nucl
PtrRHC26	POPTR_0019s02620	XM_006370949.1	225	24,166.4	6.57	chlo
PtrRHC27	POPTR_0006s20470	XM_002309278.2	202	23,031.3	9.44	nucl
PtrRHC28	POPTR_0002s08140	XM_002300935.2	210	23,956.8	4.03	nucl
PtrRHC29	POPTR_0012s08100	XM_002318002.2	391	44,217.0	7.13	plas
PtrRHC30	POPTR_0010s20910	XM_002315180.2	276	30,420.7	6.23	plas
PtrRHC31	POPTR_0008s06290	XM_002312098.2	203	32,233.9	4.84	extr
PtrRHC32	POPTR_0018s03740	XM_006371736.1	543	58,911.0	8.55	nucl
PtrRHC33	POPTR_0002s21740	XM_002302821.2	180	19,999.7	5.96	chlo
PtrRHC34	POPTR_0016s05880	XM_002322692.2	446	49,739.0	5.38	nucl
PtrRHC35	POPTR_0002s23000	XM_006386784.1	293	32,844.8	6.64	cyto
PtrRHC36	POPTR_0001s31170	XM_002298695.2	1,103	122,807.3	5.27	plas
PtrRHC37	POPTR_0001s06960	XM_002299146.1	164	17,051.7	7.54	extr
PtrRHC38	POPTR_0003s17510	XM_002304667.2	753	83,496.3	6.57	nucl
PtrRHC39	POPTR_0007s04050	XM_002310479.1	231	26,438.4	6.20	cyto
PtrRHC40	POPTR_0004s14500	XM_006384318.1	465	52,975.7	6.33	nucl
PtrRHC41	POPTR_0005s17460	XM_002307320.1	417	46,879.7	8.73	chlo
PtrRHC42	POPTR_0008s20990	XM_002312722.2	269	31,259.6	6.05	cyto
PtrRHC43	POPTR_0006s26180	XM_002309544.2	300	34,860.9	6.14	cyto
PtrRHC44	POPTR_0002s10220	XM_002301041.2	389	43,501.0	5.11	nucl
PtrRHC45	POPTR_0008s05830	XM_002312071.2	474	51,311.3	6.39	chlo
PtrRHC46	POPTR_0014s05250	XM_006375141.1	349	37,428.7	6.32	cyto
PtrRHC47	POPTR_0010s17630	XM_002315016.2	280	30,335.0	4.67	nucl
PtrRHC48	POPTR_0008s00640	XM_002310892.2	259	29,624.1	5.63	plas
PtrRHC49	POPTR_0009s09220	XM_002313138.2	330	36,360.7	8.75	chlo

Table 1 (continued)

Gene name	Accession number	NCBI locus ID	Length (aa)	MW (Da)	pI	Localization[a]
PtrRHC50	POPTR_0003s20780	XM_002303943.2	388	43,061.2	4.85	nucl
PtrRHC51	POPTR_0008s08590	XM_002312209.2	292	31,382.9	4.92	extr
PtrRHC52	POPTR_0002s20690	XM_002302672.2	320	36,980.0	8.52	nucl
PtrRHC53	POPTR_0013s01710	XM_006375682.1	432	46,630.9	6.18	nucl
PtrRHC54	POPTR_0010s05720	XM_002314558.2	133	14,885.9	5.18	cyto
PtrRHC55	POPTR_0001s12980	XM_002299318.2	638	71,492.2	8.29	nucl
PtrRHC56	POPTR_0010s17540	XM_002316097.1	218	24,635.5	4.88	cyto
PtrRHC57	POPTR_0001s31710	XM_002298673.2	225	24,150.7	8.22	chlo
PtrRHC58	POPTR_0002s00780	XM_002301867.2	200	20,805.8	8.53	chlo
PtrRHC59	POPTR_0001s09630	XM_002297987.2	212	23,036.6	4.52	chlo
PtrRHC60	POPTR_0002s12590	XM_002301136.2	765	82,596.1	6.44	nucl
PtrRHC61	POPTR_0002s08630	XM_002302210.2	365	42,012.3	6.29	chlo
PtrRHC62	POPTR_0016s14320	XM_002323037.2	189	20,323.2	6.64	cyto
PtrRHC63	POPTR_0009s01110	XM_002314268.2	291	33,239.9	6.58	cyto
PtrRHC64	POPTR_0010s25940	XM_006378811.1	259	29,816.9	5.28	cyto
PtrRHC65	POPTR_0009s05000	XM_002313348.2	243	28,008.8	8.28	nucl
PtrRHC66	POPTR_0002s04080	XM_002300745.1	344	39,671.9	8.32	chlo
PtrRHC67	POPTR_0018s01110	XM_002324797.2	434	47,462.8	6.47	plas
PtrRHC68	POPTR_0001s00610	XM_002299084.2	587	65,484.6	6.79	plas
PtrRHC69	POPTR_0003s10880	XM_002303465.2	581	64,517.6	6.15	extr
PtrRHC70	POPTR_0010s08340	XM_002314599.2	355	39,524.7	8.00	chlo
PtrRHC71	POPTR_0016s06820	XM_002323337.2	167	18,061.7	7.56	extr
PtrRHC72	POPTR_0006s26310	XM_002309552.2	429	46,775.8	6.47	plas
PtrRHC73	POPTR_0002s15660	XM_002301299.2	378	42,226.7	5.75	plas
PtrRHC74	POPTR_0008s13440	XM_002312428.2	351	39,664.9	5.80	plas
PtrRHC75	POPTR_0005s24050	XM_002306785.2	436	48,941.9	9.41	nucl
PtrRHC76	POPTR_0002s04500	XM_002302036.2	436	49,063.1	9.50	chlo
PtrRHC77	POPTR_0013s13580	XM_006376440.1	263	29,636.5	8.41	nucl
PtrRHC78	POPTR_0019s13210	XM_006371504.1	263	29,508.6	8.43	nucl
PtrRHC79	POPTR_0003s09230	XM_002304304.1	267	30,431.3	5.81	chlo
PtrRHC80	POPTR_0014s08670	XM_002320126.2	205	22,894.1	6.80	chlo
PtrRHC81	POPTR_0006s19080	XM_002309347.2	491	54,700.1	8.17	ER
PtrRHC82	POPTR_0018s10760	XM_002324464.2	466	51,840.9	6.47	nucl
PtrRHC83	POPTR_0001s16210	XM_006369012.1	276	30,653.9	5.61	cyto
PtrRHC84	POPTR_0016s02910	XM_002323174.2	308	34,457.9	6.91	nucl
PtrRHC85	POPTR_0006s03070	XM_002308820.2	308	34,449.9	6.72	nucl
PtrRHC86	POPTR_0008s22800	XM_006380133.1	289	33,634.8	9.79	nucl
PtrRHC87	POPTR_0003s18290	XM_002304701.2	195	21,220.1	6.04	nucl
PtrRHC88	POPTR_0014s03990	XM_006374994.1	228	24,642.8	8.13	nucl
PtrRHC89	POPTR_0002s13410	XM_002302434.2	227	24,825.1	8.13	nucl
PtrRHC90	POPTR_0001s22130	XM_002299621.2	386	41,789.0	5.48	nucl
PtrRHC91	POPTR_0014s18890	XM_002321252.2	213	23,481.6	9.39	nucl

[a] *plas* plasma membrane; *nucl* nucleus; *chlo* chloroplast; *extr* extracellular matrix; *cyto* cytoplasm; *ER* endoplasmic reticulum

an unrooted phylogenetic tree was constructed from alignments of the full-length C3HC4-type RING finger protein sequences (Fig. 1). The 91 C3HC4-type RING finger genes were classified into seven groups (I, II, III, IV, V, VI, and VII) containing 17, 9, 11, 12, 24, 10, and 8 members, respectively.

Almost 8,000 pairs of paralogous genes are present in the *Populus* genome (Guo et al. 2009). On the basis of the phylogenetic analysis, we identified 17 paralogous pairs among the 91 *Populus* C3HC4-type RING finger genes. Further, we used PSORT analysis to examine the protein subcellular localization of the 17 C3HC4-type RING finger gene pairs that showed close phylogenetic relationships (Fig. 1, Table 1). Among the 17 gene pairs, 11 belong to the same group, and identical subcellular localizations were predicted for their proteins. For example, PtrRHC21 and PtrRHC34 in group I were both located in the nucleus, and PtrRHC26 and PtrRHC57 in group VI were both located in the chloroplast. However, the other six paralogous pairs had different predicted locations between each member of the pair. For example, PtrRHC68 and PtrRHC69 were predicted to be localized in the plasma membrane and extracellular matrix, respectively, even though they were both placed in group II. PtrRHC75 and PtrRHC76 were predicted to be localized in the nucleus and chloroplast, respectively; however, both were classified in group IV. Thus, the same phylogenetic grouping based on sequence similarity did not necessarily correspond to the same subcellular localization. Therefore, homologous genes may show a distinct difference in gene function and signal transduction. These results were similar to those of previous analyses (Ma et al. 2009).

Chromosomal Location and Duplication of Populus C3HC4-type RING Finger Genes

In silico mapping of the gene loci showed that the 91 *Populus* C3HC4-type RING finger genes were distributed across all 19 linkage groups (LGs). In the currently released sequences, 89 genes were mapped to LGs, whereas two genes (*PtrRHC15* and *PtrRHC31*) remained on as yet unmapped scaffolds (Fig. 2). The 89 *Populus* C3HC4-type RING finger genes were distributed across all LGs, except for LG XV and XVII. The distribution of *Populus* C3HC4-type RING finger genes across the LGs appeared to be uneven. LG II encompassed the largest number (14) of genes followed by 10 on LG I. In contrast, only two genes were found on each of LG VI, LG VII, LG XII, and LG XIX. No substantial clustering of *Populus* C3HC4-type RING finger genes was observed, even on the LGs with high densities of C3HC4-type RING finger genes.

Previous studies revealed that the *Populus* genome has undergone at least three rounds of genome-wide duplication, followed by multiple segmental duplications, tandem duplications, and transposition events, such as retroposition and replicative transposition (Tuskan et al. 2006). In particular, the

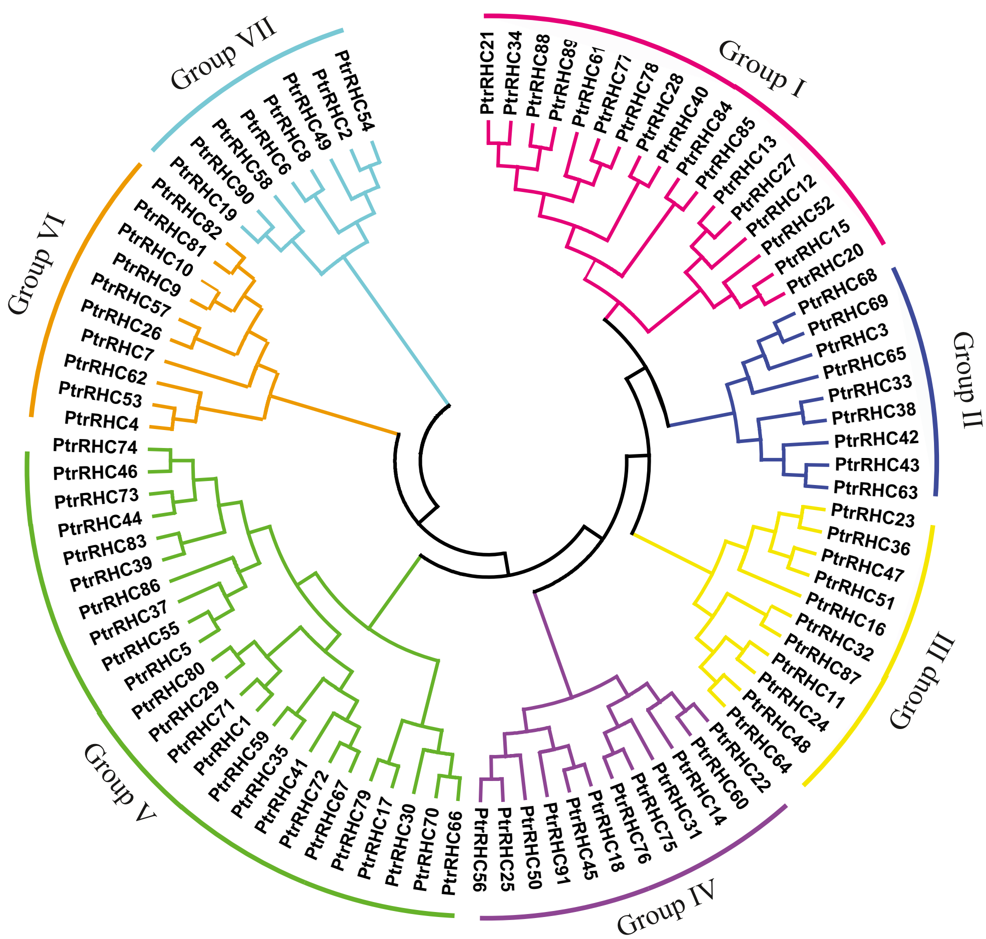


Fig. 1 Phylogenetic relationships of *Populus* C3HC4-type RING finger proteins. The deduced full-length amino acid sequences of 91 *Populus* C3HC4-type RING finger proteins were aligned using ClustalX 1.83. The phylogenetic tree was constructed with MEGA 5.0 using the neighbor-joining method. Support values from a bootstrap analysis with 1,000 replicates are specified at each node. Each subfamily is indicated by a specific *color*

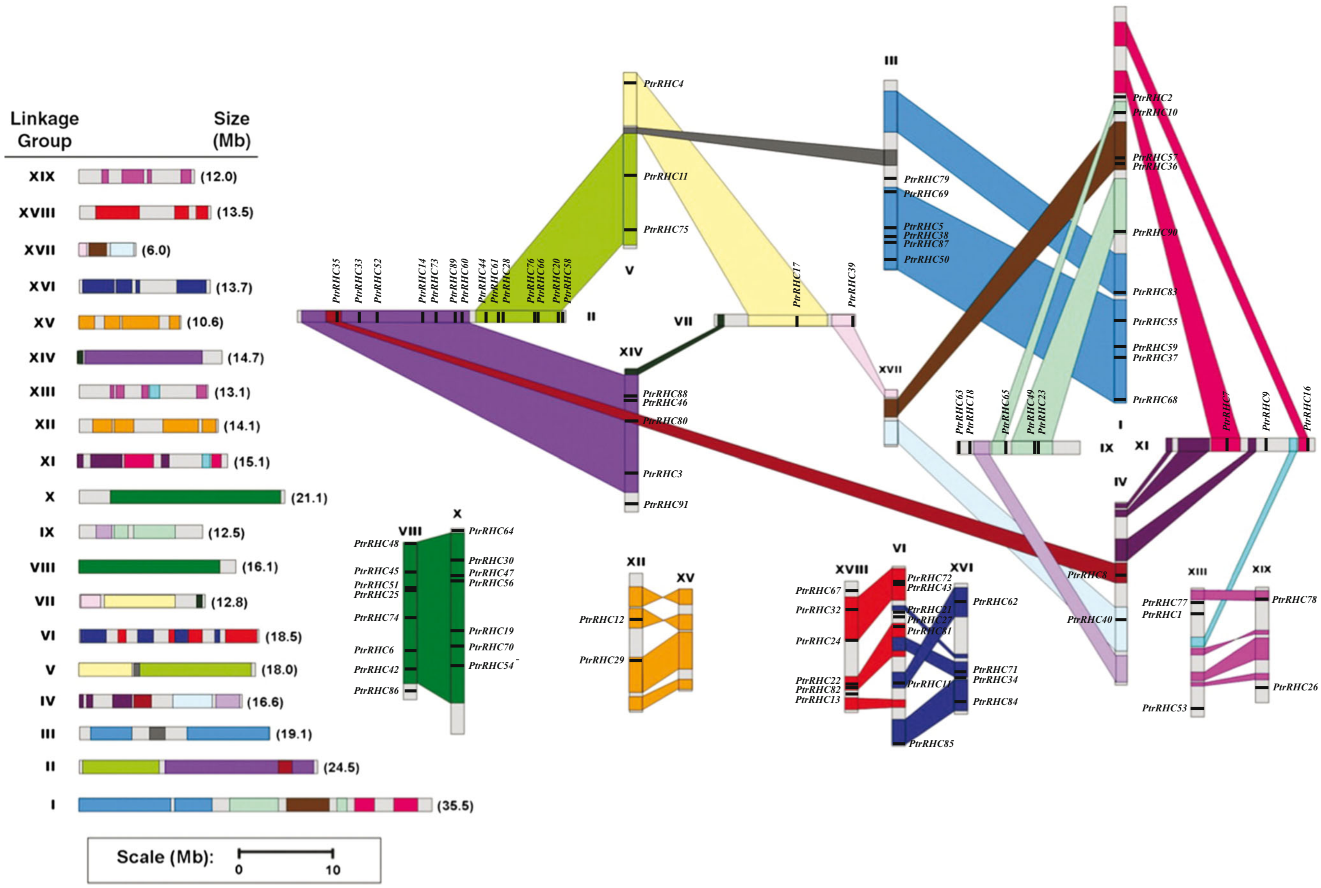
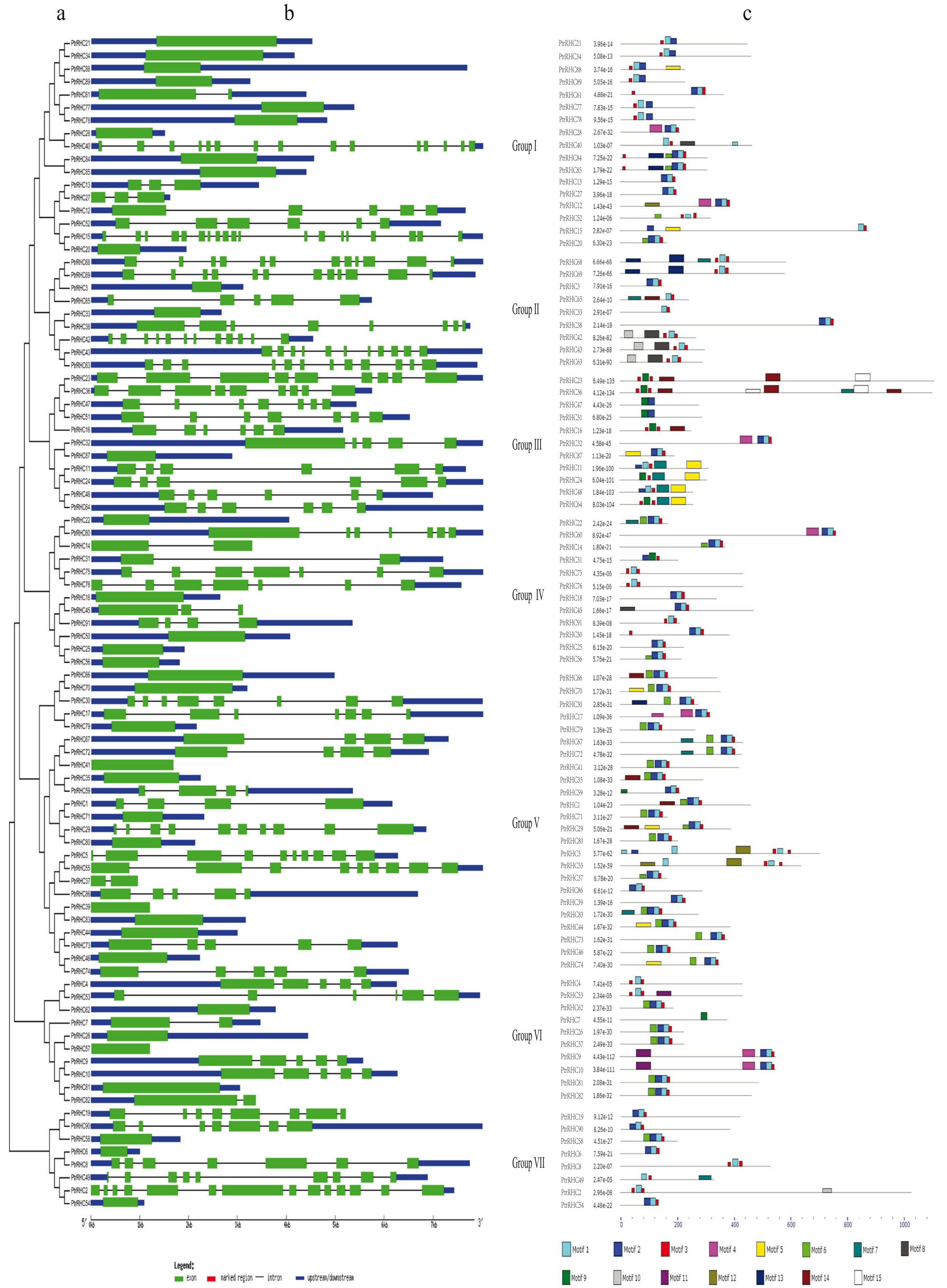


Fig. 2 Chromosomal locations of *Populus* C3HC4-type RING finger genes. Eighty-nine C3HC4-type RING finger genes were mapped to 19 linkage groups (LG), while two genes reside on unassembled scaffolds. A schematic view of chromosome reorganization due to recent whole-genome duplication in *Populus* is shown (adapted from Tuskan et al. 2006). Segmental duplicated homologous blocks are indicated by the same *color*. The *scale* represents mega bases (Mb). The LG numbers are indicated above each *bar*

segmental duplication associated with the salicoid duplication event that occurred 65 million years ago contributed significantly to the expansion of many multi-gene families (Barakat et al. 2009; Wilkins et al. 2009; Wang et al. 2013). To determine the possible relationship between the C3HC4-type RING finger genes and potential segmental duplications, we mapped *Populus* C3HC4-type RING finger genes to the duplicated blocks established in previous studies (Tuskan et al. 2006). The distribution of C3HC4-type RING finger genes relative to the corresponding duplicate blocks is illustrated in Fig. 2. In the present study, about 69 % (68 of 89) of *Populus* C3HC4-type RING finger genes were preferentially retained duplicates located in both duplicated regions. Six duplicated blocks only contained C3HC4-type RING finger genes on one of the blocks and lacked duplicates on the corresponding block. These results suggested that dynamic rearrangements might have occurred following the segmental duplication, which led to loss of some of the genes. In contrast, 17 C3HC4-type RING finger genes were located outside of any duplicated blocks. Analysis of C3HC4-type RING finger paralogous pairs showed that seven paralogous pairs remained in conserved positions on segmental duplicated blocks (Fig. 2), which suggested that these seven paralogous pairs may be derived from a segmental duplication event during the biological evolutionary process. Among the non-genome-duplicated gene pairs (*PtrRHC21/34*, *PtrRHC77/78*, *PtrRHC43/63*, *PtrRHC48/64*, *PtrRHC67/72*, *PtrRHC4/53*, *PtrRHC26/57*, *PtrRHC9/10*, *PtrRHC23/36*, and *PtrRHC11/87*), eight genes (*PtrRHC4*, *10*, *34*, *43*, *48*, *57*, *72*, and *78*) were located on duplicated segments whereas their counterparts were not on any duplicated blocks, and two genes (*PtrRHC23* and *11*) were located on one duplicated segment whereas their counterparts were on another duplicated segment. Although they were homologous genes, their real paralogous counterparts appeared to have been lost from the genome. Similar results have been reported in analyses of other *Populus* gene families (Chai et al. 2012; Hu et al. 2012). Taken together, the present results suggested that the diverse duplication events contributed to the complexity of the C3HC4-type RING finger gene family in the *Populus* genome.


a
b
c
Group I
Group II
Group III
Group IV
Group V
Group VI
Group VII
5'
3'
0 200 400 600 800 1000
Legend:
exon
marked region
intron
upstream/downstream
Motif 1
Motif 2
Motif 3
Motif 4
Motif 5
Motif 6
Motif 7
Motif 8
Motif 9
Motif 10
Motif 11
Motif 12
Motif 13
Motif 14
Motif 15

◀ **Fig. 3** **a**–**c** Phylogenetic relationships, gene structure and motif compositions of *Populus* C3HC4-type RING finger genes. **a** Multiple alignment of 91 full-length amino acid sequences of *Populus* C3HC4-type RING finger genes executed with ClustalX 1.83. The phylogenetic tree was constructed with MEGA 5.0 using the neighbor-joining method. Support values from a bootstrap analysis with 1,000 replicates are specified at each node. **b** Exon/intron structure of *Populus* C3HC4-type RING finger genes. Exon/intron structures were obtained from the Gene Structure Display Server. Exons and introns are represented by *green boxes* and *black lines*, respectively. **c** Schematic representation of the conserved motifs elucidated by MEME. Each motif is represented by a *number* in the *colored box*. *Black lines* Non-conserved sequences

Gene Structure and Conserved Motifs

Gene structural diversity and conserved motifs divergence are possible mechanisms explaining the evolution of multigene families. To clarify the present results, we first constructed a phylogenetic tree using the *Populus* C3HC4-type RING finger protein sequences, which were grouped into seven subgroups in accordance with the above-mentioned phylogenetic tree (Fig. 3a). To gain further insights into the structural diversity of *Populus* C3HC4-type RING finger genes, we analyzed the exon/intron organization in the full-length cDNA with their corresponding genomic DNA sequences of individual C3HC4-type RING finger genes in *Populus* (Fig. 3b). The coding sequences of most of the C3HC4-type RING finger genes were disrupted by introns, with numbers varying from 1 to 18. According to the number of introns, the genes were classified into three groups: a group lacking introns (36 genes), an intron-poor group (10 genes, each with 1–2 introns) and an intron-rich group (45 genes, the number of introns varying from 3 to 18).

Putative motifs were predicted using the program MEME (Fig. 3c) to further reveal the diversification of C3HC4-type RING finger genes in *Populus*. Fifteen distinct motifs were identified (Table 2). In addition, 83 genes contained motif 1 and 66 genes contained motif 2. In contrast, only two genes contained motif 10 (PtrRHC9 and 10), motif 11 (PtrRHC5 and 55), motif 13 (PtrRHC23 and 36) and motif 11 (PtrRHC4 and 53), which suggested that these genes may have special functions.

The similar gene structures and conserved motifs of C3HC4-type RING finger genes in the same subfamilies may provide additional support for the phylogenetic analysis. Conversely, the differences in gene structure and motif composition among different subfamilies indicated that they might be functionally divergent.

Analysis of Promoter cis-Elements

Cis-elements play key roles in the transcriptional regulation of genes that control abiotic stress responses, such as drought and abscisic acid (ABA). To identify putative cis-acting regulatory DNA elements in C3HC4-type RING finger genes, promoter sequences (−2, 000 bp) were obtained from the NCBI database, and the cis-elements of these 91 promoters were examined using the PlantCARE database (Table S3). Many abiotic stress-related cis-elements, including ABA-responsive elements (ABREs) (Narusaka et al. 2003), heat stress-responsive elements (HSEs) (Díaz-Martín et al. 2005), dehydration-responsive elements (Shinwari et al. 1998), and GCC-boxes (Song et al. 2005), have been identified. All of these cis-elements were observed in the present study. The majority of C3HC4-type RING finger genes contained HSE, G-box, and ABRE motifs. In addition, 39 genes also contained a W-box, which is found in the promoter regions of plant defense-related genes and has been correlated with drought and ABA stress (Singh et al. 2002).

Table 2 Motif sequences of C3HC4-type RING finger genes identified in *P. trichocarpa*

Motif	Width (aa)	Best possible match
1	18	CGHFFHWDCIDKWLKQHN
2	21	DTECAICLEEYEPGEEVRELP
3	9	TCPICRKEL
4	41	HHRDMRLDIDDMSYEELLALGERIGNVSTGLSEDTISSCLK
5	50	EHCYGFYTMFVIAFVIVGIAYGFFIATMCGQKIWQRHYHILAKQELTKEY
6	21	RGLDQKVIKALPVYKYKKIKN
7	42	DNWWRKMKFRFFVTRDHVFIFIVVQLVIAFMGGFVYKMYGDG
8	50	GVCMGKYFCEKCKFFDDDTSKKQYHCDDCGICRIGGKENFFHCYKCGCCY
9	21	PCACKGSLKYAHRKCVQHWFN
10	50	DAHLVDRYDGAMFYGMPQYHGVHPHRQYHGPNLDLSVATAPNFYVPYMTP
11	50	YAPETGVRYDGVYRIEKCWRKNGIQGFKVCRYLFVRCDNEPAPWTSDVQG
12	15	PPPASKAAIEAMPTV
13	41	FQEFVVGMAMKTCHVLQFFLRLSFVLSVWLLIIPFITFWIW
14	41	WLYANGCRSLPEFTMDDWAHDEDLYDLSYSEMSFGVHWCPF
15	29	FACICYAMPFIICITICCCLPCIIAILYY

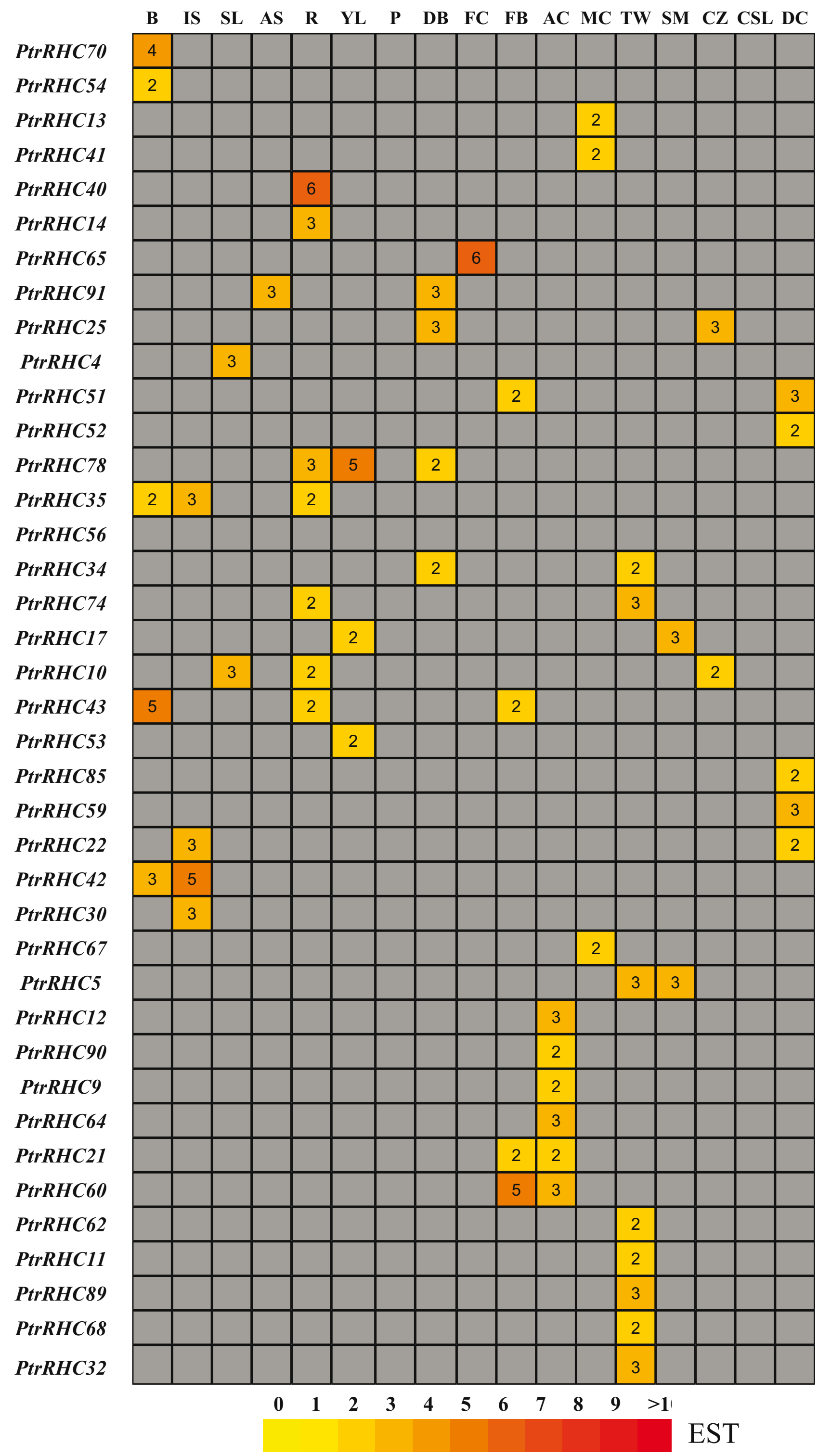


Fig. 4 In silico expressed sequence tag (EST) analysis of *Populus* C3HC4-type RING finger genes. EST frequency for each gene was calculated by evaluating its EST representation among 17 cDNA libraries accessible with the exNorthern tool at PopGenIE. The *color bar* at the bottom represents the frequencies of EST counts. *B* Bark, *IS* imbibed seeds, *SL* senescing leaves, *AS* apical shoot, *R* roots, *YL* young leaves, *P* petioles, *DB* dormant buds, *FC* female catkins, *FB* flower buds, *AC* active cambium, *MC* male catkins, *TW* tension wood, *SM* shoot meristem, *CZ* cambial zone, *CSL* cold-stressed leaves, *DC* dormant cambium

Differential Expression Profile of Populus C3HC4-type RING Finger Genes

Publicly available ESTs are considered a useful means of studying gene expression profiles using the Digital Northern tool (Ohlrogge and Benning 2000). We first carried out a preliminary analysis of C3HC4-type RING finger gene expression across different tissues and organs by counting the frequencies of ESTs in different *Populus* cDNA libraries (Fig. 4). A complete search of the digital expression profiles in PopGenIE yielded 39 *Populus* C3HC4-type RING finger genes in the cDNA libraries.

Among the 39 genes, 5 showed the highest transcript accumulation in bark, 4 in imbibed seeds, 2 in senescing leaves, 1 in the apical shoot, 7 in roots, 3 in young leaves, 4 in dormant buds, 1 in female catkins, 4 in flower buds, 6 in active cambium, 3 in male catkins, 8 in tension wood, 2 in the shoot meristem, 2 in the cambial zone, and 5 in dormant cambium. By contrast, no gene expression in petioles and cold-stressed leaves was indicated in the EST libraries. However, the genes might be expressed in other libraries for which data is not available.

Usually, high or preferential expression in tissues or organs suggests that a gene might play an important role (Liu et al. 2013, 2014). *PtrRHC65* showed a notably high expression level in female catkins and *PtrRHC60* showed high expression in flower buds, which indicated that these genes show a close relationship with flower differentiation and floral organ formation. *PtrRHC17* showed a high expression level in young leaves and the shoot meristem, which suggested that it may play an important role during reproduction. *PtrRHC5*, *32*, *74*, and *89* were highly expressed in tension wood, which indicated that these genes are related to secondary wall biosynthesis. *Populus*, as a woody tree, produces abundant wood compared with herbaceous plants. Wood (a secondary tissue) is generated by a succession of five major steps, consisting of cell division, cell expansion, cell wall thickening (involving biosynthesis and deposition of cellulose, hemicellulose, cell wall proteins, and lignin), programmed cell death, and heartwood formation (Plomion et al. 2001). In many angiosperm trees, including *Populus*, secondary wall development is modified greatly when tension wood is formed as a gravitational response to stem movements caused, for example, by wind or load (Hellgren et al. 2004; Pilate et al. 2004). Tension wood is characterized mainly by abnormal fibers, which are poorly lignified and have an additional thick layer in the secondary cell wall (Lafarguette et al. 2004). Therefore, tension wood can be involved in the formation of wood. In consideration of these factors and the high expression level of C3HC4-type RING finger genes in tension wood, we suggest that the C3HC4-type RING finger gene family might contribute to wood formation. This conclusion is consistent with previous research (Andersson-Gunnerås et al. 2006). In summary, the expression profiles demonstrated that the C3HC4-type RING finger genes show a broad expression pattern across different tissues and organs. These tissue- and organ- preferentially expressed genes may play important roles in the regulation of plant growth and development.

	Fungi	Fungi	Defence	Defence	Defence	Defence	Defence	Defence	Defence	Defence	Defence	Habituation TA
PtrRHC26	3.37	3.4	3.26	3.08	3.14	3.45	3.71	3.53	3.41	3.24	3.22	3.65
PtrRHC22	2.5	2.8	2.5	2.6	2.5	2.7	2.7	2.4	2.7	2.6	2.7	3.2
PtrRHC13	9.2	9.3	8.8	9.1	8.7	8.8	8.6	8.9	9	8.9	8.6	6.7
PtrRHC32	7.1	7.2	8.2	8.2	7.9	8.3	8	8.2	8.4	8.3	7.9	8.5
PtrRHC24	6.2	6.2	9.1	9.1	8.8	9	8.9	9.3	9.4	9.2	9	4.2
PtrRHC34	7.7	7.4	7.5	7.4	7.7	7.4	7.6	7.5	7.5	7.6	7.6	8.1
PtrRHC3	11.2	10.7	7.2	7.1	7	7	6.8	7	6.8	6.9	7	6.5
PtrRHC46	3.3	3	2.9	2.9	2.8	3	2.9	3.1	3	2.7	2.7	2.9
PtrRHC29	7.2	7.3	7.2	7.2	7.4	7.5	7.4	7.2	6.8	7.1	7.4	7.7
PtrRHC1	9.7	9.5	8.5	8.2	8.6	8.5	8.6	8.6	8.2	8.5	8.7	8.8
PtrRHC12	7.1	7.5	7.5	7.5	7.8	7.7	7.8	7.6	7.5	7.5	7.8	11.1
PtrRHC16	7.3	7.3	7.2	7.5	7.1	7.3	7.2	7.3	7.3	7.1	7.1	7.8
PtrRHC9	6.9	6.9	7	6.8	6.9	7.2	7	7	7	7	7.1	6.3
PtrRHC30	7.6	7.7	8.2	8.4	8.3	8.3	8.3	8.2	8.5	8.4	8.2	8.9
PtrRHC7	4.2	4.3	4.1	3.8	3.8	4.2	3.8	3.6	3.3	3.5	3.9	4.8
PtrRHC47	6.2	6.1	6.1	6.3	6.4	5.9	6.4	6.3	6.5	6.4	6	5.2
PtrRHC19	3	3.2	3.3	3.5	3.3	3.5	3.5	3.5	4.1	3.5	3.9	4
PtrRHC23	8.1	7.7	7.1	6.6	7.4	6.8	7.8	6.6	6.9	7	6.8	7.5
PtrRHC49	3.6	3.6	3	2.8	2.9	2.9	2.9	3	3	3.1	3	6.9
PtrRHC18	4.7	4.7	3	3.6	3.2	3	3.3	3.2	3.4	3.2	3.1	6.7
PtrRHC42	4.7	4.8	7.1	6.9	7.1	7.1	6.8	6.7	6.8	7	7.1	8.3
PtrRHC25	9.8	10	11.4	11.4	11.3	11.2	10.9	11	11.2	11.2	10.8	7.5
PtrRHC45	7.2	7.3	7.4	7.3	7.5	7.5	7.3	7.4	7.3	7.4	7.7	6.2
PtrRHC48	7.5	7.6	7.3	7.6	7.1	7.1	6.6	6.7	7	6.9	6.9	7.9
PtrRHC17	8.5	8.7	3	2.8	2.8	3.2	2.8	3.1	2.8	2.7	2.9	8.9
PtrRHC39	2.3	2.4	2.4	2.4	2.3	2.4	2.3	2.3	2.5	2.4	2.5	3.8
PtrRHC21	7.3	7.2	6.7	6.5	6.9	6.3	6.9	6.4	6.7	6.7	6.5	7.7
PtrRHC43	9.6	9.2	4.8	4.9	4.7	4.6	4.9	4.1	4.7	5.3	4.8	6.8
PtrRHC11	8.3	8.2	7.8	8.1	7.8	7.5	8	7.7	8.3	8.1	7.8	6.8
PtrRHC41	2.6	2.7	3	2.8	2.6	2.9	2.8	2.8	2.8	2.7	2.9	6.6
PtrRHC8	2.6	2.7	2.6	3	2.6	2.8	2.9	2.8	2.9	2.9	3	5.5
PtrRHC40	5.9	6.1	6.9	7.1	7	7	6.8	6.8	6.8	6.9	6.9	6.8
PtrRHC38	7.7	7.6	6.8	6.9	6.8	6.8	6.9	6.7	7	7	7	8.5
PtrRHC5	2.8	3	3.5	3.4	3.4	3.1	3.3	2.9	3.2	3.4	3.3	6.8
PtrRHC35	7.5	7.8	8.5	8.7	8.2	8.3	8.3	8.6	8.3	8.5	8.2	8.5
PtrRHC33	2.1	2.3	2.2	2.3	2.3	2.5	2.2	2.3	2.3	2.2	2	9.2
PtrRHC14	2.4	2.5	3	2.7	3.1	3.1	3.2	3	2.7	3	3	3.1
PtrRHC44	3.6	4	3.6	3.2	3.5	3.5	3.5	3.3	4.1	3.5	3.6	2.7
PtrRHC28	4	4.6	3.4	3.4	3.3	3.8	3.2	3.3	3.5	3.1	3.7	5.7
PtrRHC20	5	5.1	3.1	2.9	2.8	3.2	3.2	3	3	3	3	5.5
PtrRHC10	6.8	6.5	6.9	6.7	7.7	6.9	8	6.3	7.2	7.1	7.2	7.4
PtrRHC2	2.8	2.9	2.5	2.5	2.5	2.3	2.6	3	2.5	2.6	2.7	4.6
PtrRHC37	4.6	4.2	5	5	5.1	5.2	5	5.2	4.9	4.6	5.3	4
PtrRHC82	4.3	4.6	4.4	4.6	4.9	5.2	4.9	5	4.2	4.6	4.9	2.8
PtrRHC78	4.1	4	4.3	3.8	5	4.3	4.5	3.4	4	4.3	4.7	2.4
PtrRHC84	8.5	8.3	7	7.2	7.4	7	7.6	7.1	7.6	7.3	7.2	7.9
PtrRHC71	2.7	2.6	2.8	2.3	2.7	2.5	2.8	2.8	2.7	2.5	2.6	4.9
PtrRHC62	4.8	4.4	3.3	4.2	3.5	4.1	3.5	4	4	4.3	3.7	3
PtrRHC67	6.1	5.8	5.7	6.1	5.9	5.6	5.9	5.3	6.3	6.1	5.5	7
PtrRHC91	8.9	9	8.6	8.8	8.5	8.7	8.5	8.7	8.9	8.6	8.4	8.8
PtrRHC88	3.8	3.6	3.1	3.2	3	3.1	2.8	3.2	3.2	3	3.1	3.7
PtrRHC80	6	5.9	3.8	4.1	4	4.4	4.6	4.2	3.9	3.7	4.3	4.2
PtrRHC77	7.6	7.8	10.3	9.9	10.9	10.2	11	10.1	10.2	10.5	11	8.1
PtrRHC53	8.7	8.7	8.3	8.3	8.2	8.5	8.5	8.6	8.4	8.5	8.6	8.3
PtrRHC64	8.2	8.6	7.4	7.5	7.1	7.3	7.2	7.3	7.5	7.4	7.3	10.2
PtrRHC56	7.2	7.2	7.3	7.6	7	7.2	6.7	7.3	7.2	7.3	7.2	6.8
PtrRHC70	2.4	2.4	2.9	3	2.9	3.1	2.7	2.8	2.7	2.8	2.7	6.9
PtrRHC54	4.2	3.9	3.7	3.8	3.7	3.9	3.6	3.8	3.7	3.7	3.5	4.3
PtrRHC63	7	8.3	8.6	8.8	8.9	8.8	9.1	8.8	8.5	9	9.3	6
PtrRHC65	7.3	7.5	7.2	7.2	7.9	7.6	8	7.3	6.5	6.7	7.2	6.4
PtrRHC86	3.1	2.9	2.4	2.5	2.8	2.6	2.8	2.6	2.6	2.8	2.7	3.9
PtrRHC74	4.6	4.5	3.8	3.8	3.9	3.9	4.1	4.1	3.7	3.9	3.7	5.1
PtrRHC51	8.6	8.7	6.7	7	6.8	7.1	7.5	7.1	7.7	7.5	7.3	10.1
PtrRHC72	7.3	7.4	7.5	7.3	7.5	7.7	7.2	7.4	7.6	7.5	7.4	7.3
PtrRHC81	2.8	3	2.3	2.5	2.3	2.2	2.4	2.6	2.4	2.5	2.1	2.8
PtrRHC85	9.5	9.5	9.1	9.1	9.2	9.3	9.1	9.1	9.2	9.1	9.1	9.5
PtrRHC75	4.4	4.6	6.5	6.3	6.1	6.8	6	6.5	6.3	6.5	6.4	3.7
PtrRHC50	7.7	7.8	7.7	7.8	8.2	8	8.6	7.9	7.9	7.8	8	7.1
PtrRHC87	8.6	8.5	9	9.1	8.8	9	9	9.2	8.7	9.1	8.9	9.1
PtrRHC69	3.9	4.2	2.2	2.3	2.2	2.5	2.3	2.6	2.4	2.2	2.4	3.2
PtrRHC79	2.1	2.2	2.2	2.4	2.3	2.3	2.2	2.5	2.3	2.3	2.4	2.5
PtrRHC73	8.1	7.8	7.9	7.8	7.7	7.8	7.7	7.8	7.5	7.6	7.7	5.1
PtrRHC52	2.7	2.9	7.5	7.1	7.5	7.3	7.8	7.2	7	7.4	7.6	3.2
PtrRHC89	6.9	6.8	6.6	6.3	6.3	6.7	6.3	6.5	6.6	6.6	6.6	6.3
PtrRHC60	7.9	7.9	7.4	7.1	7.3	7.1	7.7	7.5	7.5	7.4	7.3	8
PtrRHC61	3	2.7	3.1	3.2	3.4	3.4	3.5	3.4	3.4	3.5	3.4	3.2
PtrRHC76	5.5	5.5	4.3	4.2	4.2	4.4	4.3	4.3	4.5	4.3	4.5	6
PtrRHC66	2.3	2.4	2.7	2.3	2.9	2.5	2.8	2.3	2.4	2.7	3.2	2.5
PtrRHC58	7	6.8	7.6	7.8	7.7	7.7	8.2	7.9	8.5	8	7.7	7.8
PtrRHC57	3.1	3.1	2.6	2.3	2.5	2.5	2.5	2.5	2.4	2.4	2.3	2.6
PtrRHC90	6.3	6.1	4.8	4.5	5.3	4.7	5.5	4.5	4.7	4.9	4.9	6.9
PtrRHC83	3.5	3.2	3.5	3.8	4	3.3	4.1	3.1	3.1	3.2	3.5	4.4
PtrRHC59	7.4	7.5	6	6.1	6.9	6.1	7	6.1	6.4	6.3	6.4	5.6
PtrRHC68	8.2	8.1	7.7	7.5	7.6	7.8	7.4	7.8	7.8	7.8	7.5	8.1

magnitude of change	0	1.14	2.28	3.42	4.57	5.71	6.85	8	9.14	10.28
greater than zero										

Fig. 5 Differential expression of *Populus* C3HC4-type RING finger genes under different biotic stresses. Data were obtained with the exPlot tool in PopGenIE. The biotic stress conditions are represented by GSM sample numbers. *Yellow* and *red* indicate low expression and high expression levels, respectively

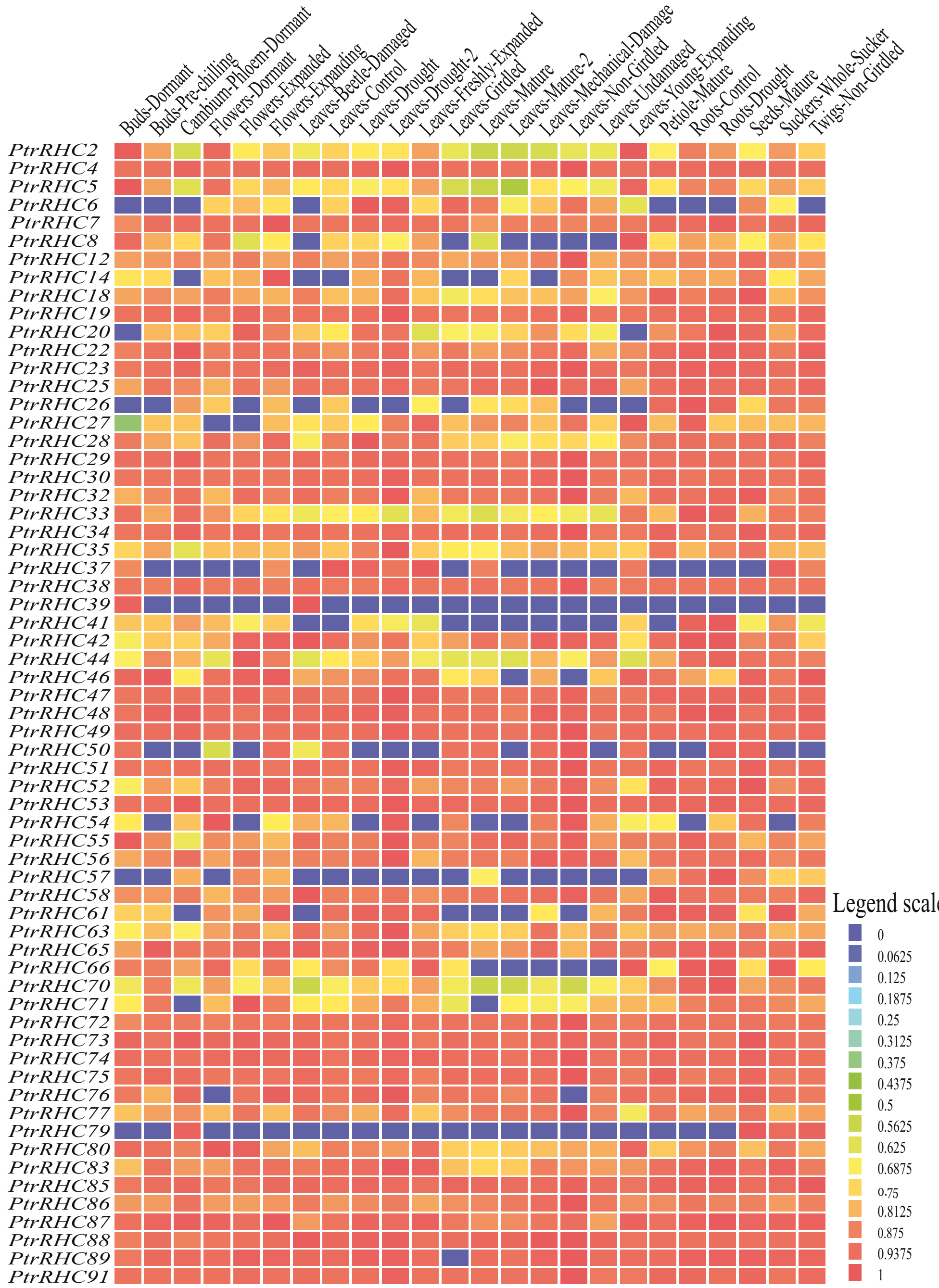


Fig. 6 Differential expression of *Populus* C3HC4-type RING finger genes across various developmental stages and under different stress conditions. The heatmap was visualized using the Heatmap tool in PopGenIE. *Blue* and *red* indicate low and high levels of transcript abundance, respectively

Additionally, the expression patterns were analyzed under fungal infection, thaxtomin A (TA) treatment, beetle damage, mechanical damage, and drought. Fungi are a primary cause of plant disease and are responsible for about 30 % of emerging diseases in plants (Anderson et al. 2004). Thaxtomin A—a phytotoxin produced by the phytopathogen *Streptomyces scabies*—can activate a cellular program leading to cell death (Brochu et al. 2010). Fungal infection and TA treatment caused upregulation of 84 C3HC4-type RING finger genes (Fig. 5). Among these genes, *PtrRHC25* and *PtrRHC85* showed remarkably high expression levels under all infection treatments. *PtrRHC12*, *33*, *51*, *64*, *85*, and *87* were highly expressed in TA-habituated cells. These results suggested a high probability of their participation in the anti-fungal signal transduction pathway and enhanced resistance for TA. Beetle damage induced upregulation of nine C3HC4-type RING finger genes (*PtrRHC18*, *20*, *35*, *39*, *58*, *63*, *65*, *77*, and *86*) and mechanical wounding induced upregulation of six genes (*PtrRHC20*, *35*, *44*, *54*, *58*, and *63*) (Fig. 6). Notably, *PtrRHC20*, *35*, and *63* were upregulated in both beetle-induced and mechanical damage. Drought is the main environmental stress that most land plants encounter during their

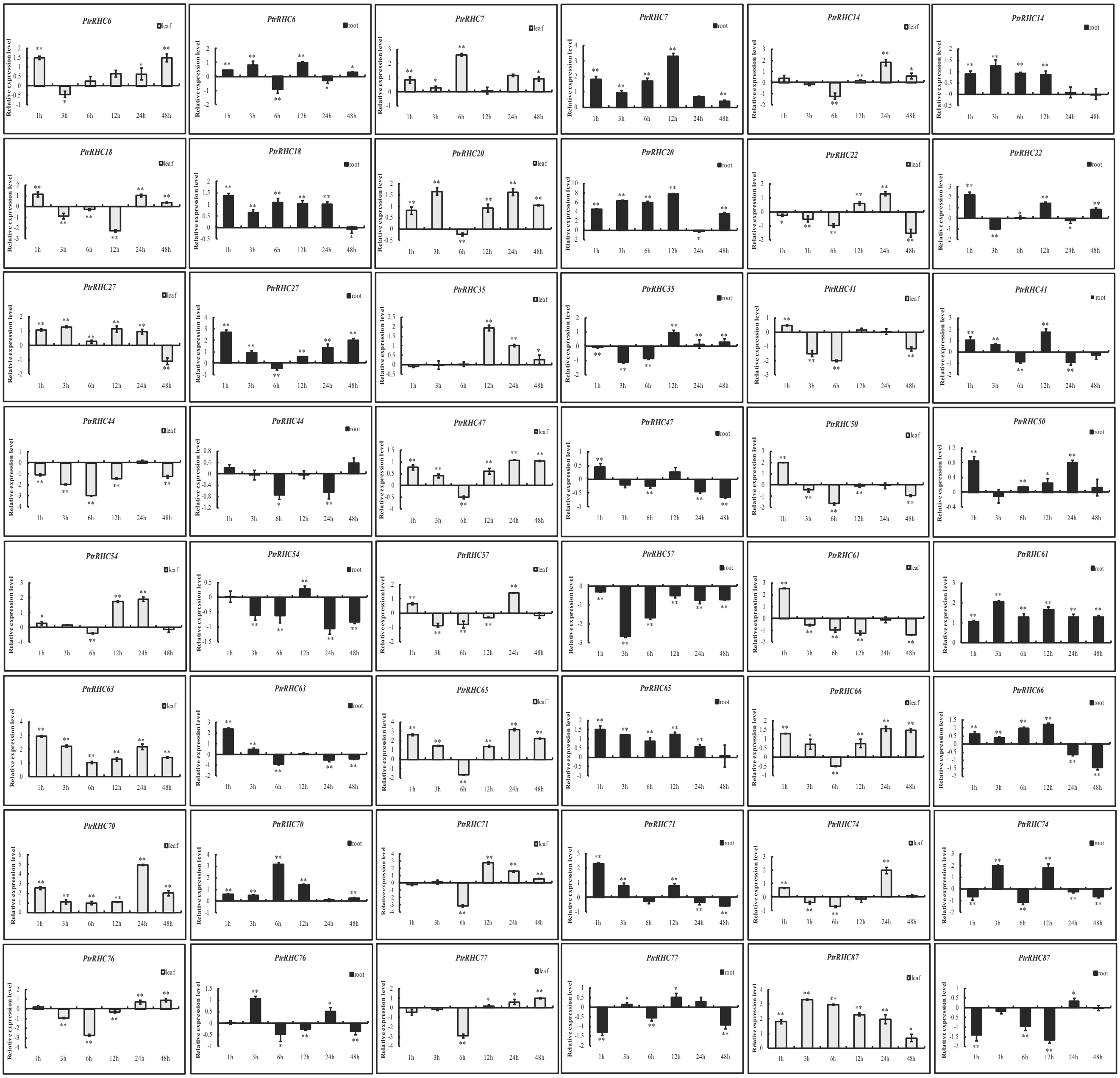


Fig. 7 Expression analysis of 24 selected C3HC4-type RING finger genes under drought stress using RT-qPCR. The relative mRNA abundance of 24 selected C3HC4-type RING finger genes was normalized with respect to reference genes. The *x*-axis represents time points after drought treatment. *Error bars* represent the standard deviation from three technical repeats. Stress treatment groups that showed a significant difference in transcript abundance compared with the control group are indicated by *asterisks* (* P<0.05, ** P<0.01)

life span. According to the present data set, 52 genes were upregulated in leaves and 39 genes were upregulated in roots under drought stress (Fig. 6). Thus, the site of activity of the gene was also different under drought stress. This result was consistent with previous analyses that revealed that the C3HC4-type RING finger gene family contributed to variation in the drought response (Cho et al. 2011; Lenka et al. 2011). Taken together, our results suggested that C3HC4-type RING finger genes play crucial roles in response to biotic and abiotic stresses.

Validation of C3HC4-type RING Finger Gene Expression by RT-qPCR

Previous studies suggested that C3HC4-type RING finger genes were closely associated with drought stress (Kim and Kim 2013; Müller et al. 2014). To verify the expression profiles of *Populus* C3HC4-type RING finger genes obtained by heatmap analysis, RT-qPCR was performed for 24 selected genes under drought stress. The results were broadly consistent with the microarray data: six genes (*PtrRHC35*, *47*, *54*, *63*, *66*, and *87*) showed the highest transcript levels in leaves and transcripts of four genes (*PtrRHC18*, *50*, *61*, and *65*) were most abundant in roots. *PtrRHC7*, *20*, and *70* not only exhibited a high expression level in leaves, but also were upregulated in roots under drought stress (Figs. 6, 7). This result suggested that these genes play an important role under water-deficient conditions. *PtrRHC57* and *74* showed their highest expression levels at 24 h after drought treatment in leaves, whereas the transcription level of *PtrRHC27* and *71* exhibited a peak at 1 h after drought treatment in roots (Fig. 7). This finding implied that these genes exert their functions at different time points. Moreover, expression patterns were not identical between family members. It is also noteworthy that all upregulated gene promoter sequences contained cis-elements related to drought stress, such as ABRE, G-box, HSE and W-box elements.

Considering all of the above evidence, we speculated from the diverse expression patterns that the C3HC4-type RING finger genes might be involved in different drought signal regulatory networks. Therefore, it would be interesting to undertake further functional studies of these C3HC4-type RING finger genes to establish the interactions of the biochemical pathways that are activated during the drought stress response.

Conclusions

Characteristics of the C3HC4-type RING finger gene family were documented initially in the model plants *Arabidopsis* and rice. However, this gene family has not been studied previously in the model tree *Populus*. In the present study, a comprehensive analysis of phylogenetic relationships, chromosomal location, gene structure, conserved motifs, cis-elements, and expression profiling of the C3HC4-type RING finger gene family in *Populus* was performed. Ninety-one full-length C3HC4-type RING finger genes in the *Populus* genome were identified, which were clustered phylogenetically into seven distinct subfamilies. The genes were distributed non-randomly across 17 LGs, and segmental duplications contributed significantly to the expansion of the *Populus* C3HC4-type RING finger gene family. Cis-elements contained in the C3HC4-type RING finger genes provide clues to their functions and expression under different biotic and abiotic stresses. We identified a subset of *Populus* C3HC4-type RING finger genes with putative functional roles in wood formation and drought response. The information obtained in the current study could help in the selection of appropriate candidate genes for further functional characterization to unravel their divergent roles.

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